PES UNIVERSITY • DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Lab 2: Agile Product Backlog & Epics Specification

Agile Requirements Transformation, User Story Mapping & Fibonacci Estimation

Student Name: Pranav
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Problem Statement #35: Customizable Subscription Box Scheduler
Domain: Retail, E-Commerce & Finance
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Project Key: SCRUM

1. Overview & Agile Transformation Methodology

In this lab, the functional requirements identified during Lab 1 for the *Customizable Subscription Box Scheduler* were transformed into Agile Epics and prioritized User Stories. Estimation was performed using the **Fibonacci sequence (1, 2, 3, 5, 8...)** via collaborative **Planning Poker** conventions, reflecting implementation complexity, technical uncertainty, and end-user business value.

2. Agile Epics & User Stories Catalog

Epic / Issue Key	User Story Title	Agile User Story Format (As a... I want... So that...)	Points	Priority	Lab 1 Trace
Epic 1: Subscription Box Customization Personalization & Swapping	**SCRUM-8** Manage Preference Tags	**As a** subscriber, **I want to** set and update preference tags (e.g., "vegan", "dark roast"), **so that** my monthly box curates items matching my taste.	5 SP	HIGH	FR-002
	SCRUM-9 Swap Box Items	**As a** subscriber, **I want to** swap individual monthly box items up to 48 hours before renewal, **so that** I receive exactly the products I desire.	5 SP	HIGH	FR-001
Epic 2: Subscription Cycle Management Schedule & Billing Controls	**SCRUM-10** Pause Delivery	**As a** subscriber, **I want to** pause recurring shipments for up to 3 months before renewal, **so that** I am not billed or sent boxes while traveling.	5 SP	HIGH	FR-001
	SCRUM-11 Skip Billing Cycle	**As a** subscriber, **I want to** skip an upcoming monthly billing cycle with auto-resumption next month, **so that** I don't need to permanently cancel.	3 SP	MEDIUM	FR-003
Epic 3: Fulfillment & Renewal Management Operations & Notifications	**SCRUM-12** View Fulfillment Manifest	**As a** fulfillment lead, **I want to** view and filter monthly box orders by category and destination, **so that** packing and inventory allocation are organized.	5 SP	HIGH	FR-004
	SCRUM-13 Export Fulfillment Manifest	**As a** fulfillment lead, **I want to** export shipping labels for 10,000 boxes in under 1 minute, **so that** automated warehouse packing has no delay.	5 SP	HIGH	FR-004, NFR-001
	SCRUM-14 Renewal Reminder Notifications	**As a** subscriber, **I want to** receive automated reminder alerts 72 hours prior to renewal date, **so that** I have sufficient time to customize, pause, or skip.	3 SP	MEDIUM	FR-005

Planning Poker Estimation Summary: Total Product Backlog Effort = **31 Story Points**. Sprint 1 was allocated **18 Story Points** (SCRUM-8, SCRUM-9, SCRUM-10, SCRUM-11) focusing on subscriber personalization and lifecycle self-service. Sprint 2 was

allocated **13 Story Points** (SCRUM-12, SCRUM-13, SCRUM-14) focusing on warehouse fulfillment operations and automated notification services.

3. Jira Epics, User Stories & Story Points Evidence

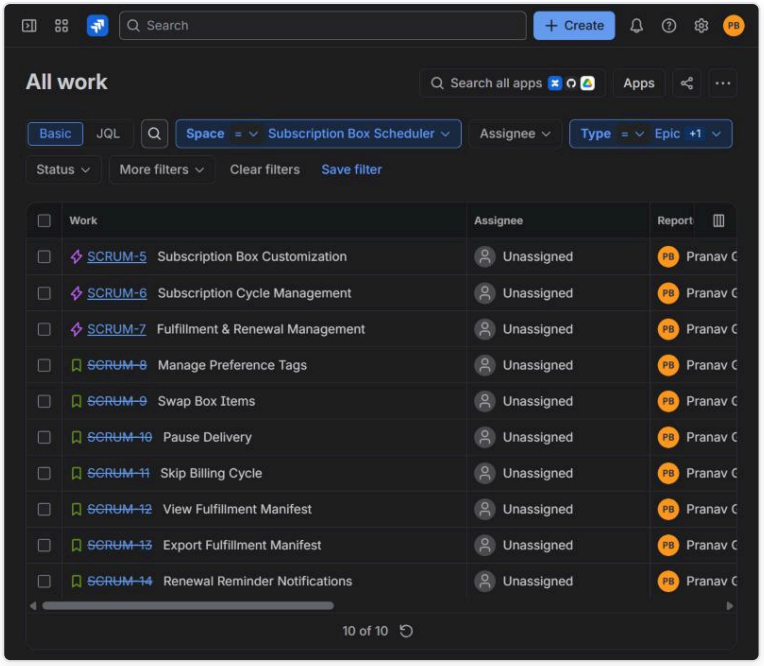


Figure 1: Jira work item search for project SCRUM showing the 3 Epics (SCRUM-5 to 7) and 7 User Stories (SCRUM-8 to 14). Both sprints are complete, so the stories no longer appear on the Backlog page.

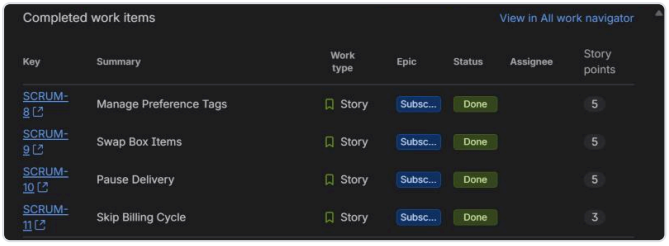


Figure 2: Story points assigned to the Sprint 1 stories (18 SP), from Jira's Sprint 1 report.

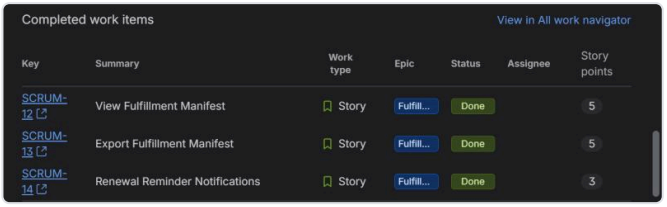


Figure 3: Story points assigned to the Sprint 2 stories (13 SP), from Jira's Sprint 2 report.

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Lab 2: Sprint Simulation, Burndown Chart & Reflection Analysis

Active Sprint Simulation (To Do → In Progress → Done), Burndown Tracking & Post-Sprint Reflection

Student Name: Pranav
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Problem Statement #35: Customizable Subscription Box Scheduler
Sprint Dates (Jira): Sprint 1: 6–20 Sep 2026 | Sprint 2: 7 Sep–4 Oct 2026
Committed Effort: Sprint 1: 18 SP | Sprint 2: 13 SP
Completion Status: 100% Completed (31 / 31 SP)

1. Active Sprint Boards (Simulation Evidence)

Two distinct sprints were planned and executed in Jira. Tasks were transitioned systematically from **To Do** → **In Progress** → **Done** adhering to the Definition of Done (DoD).

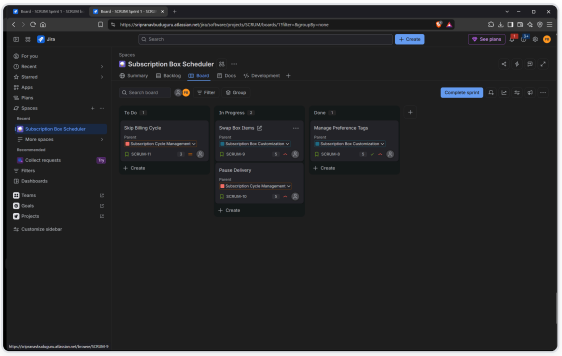


Figure 4: Sprint 1 Board during active simulation (SCRUM-8 to SCRUM-11).

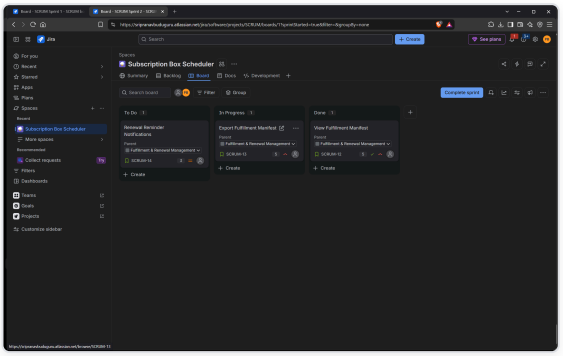


Figure 5: Sprint 2 Board progressing tasks (SCRUM-12 to SCRUM-14).

2. Burndown Charts

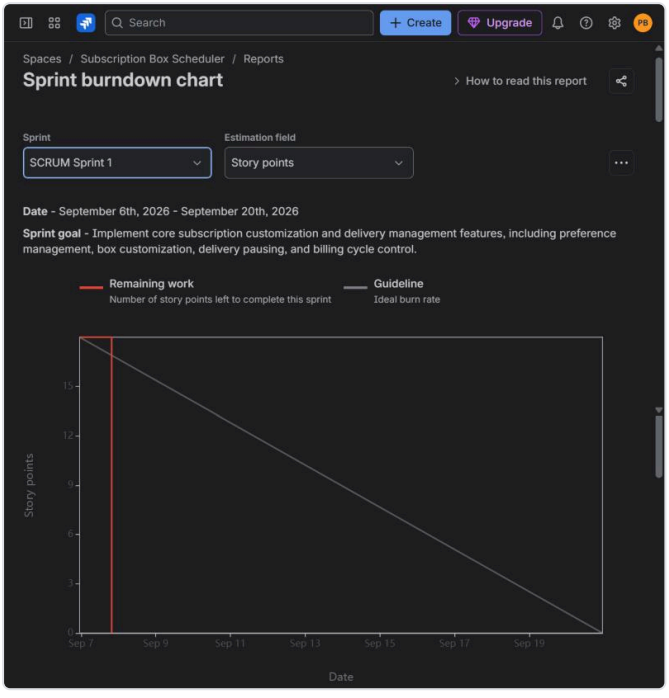


Figure 6: Jira Sprint burndown chart, Sprint 1 (18 SP).

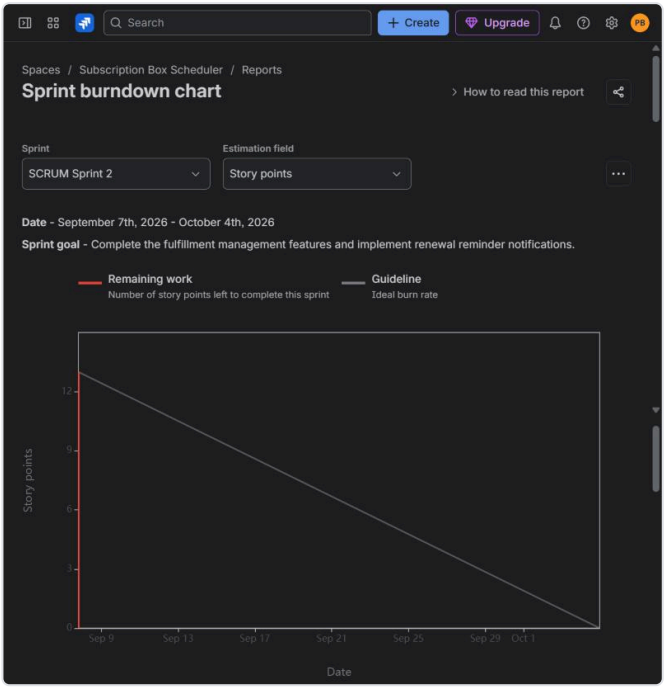


Figure 7: Jira Sprint burndown chart, Sprint 2 (13 SP).

In both charts the red *remaining work* line stays flat and then drops straight to 0. Jira records all four Sprint 1 stories as moved to Done on 7 Sep, the day after the sprint started, and all three Sprint 2 stories as moved to Done on 7 Sep, the day that sprint started. Neither sprint had scope changes, and no work items were left incomplete.

3. Answers to Handout Reflection Questions

Q1: Did your estimations reflect the actual effort?

The burndown cannot confirm it. The relative sizes still make sense: the 5-point stories (for example SCRUM-9 Swap Box Items and SCRUM-13 Export Fulfillment Manifest) carry the hardest constraints, such as the 48-hour cutoff and exporting 10,000 labels in under a minute (NFR-001), while the 3-point stories (SCRUM-11 Skip Billing Cycle and SCRUM-14 Renewal Reminder Notifications) are mostly simple state changes and scheduled alerts. However, every story was moved to Done at almost the same time, so Jira holds no record of how long each one took. Checking estimates against actual effort would need the stories to be updated as the work really progressed.

Q2: Was your backlog well-prioritized?

Yes. The subscriber-facing features (preference tags, swapping items, pausing delivery and skipping a billing cycle) were placed in **Sprint 1** so the core customer journey came first. The fulfillment features (viewing and exporting the manifest) and the 72-hour renewal reminders were placed in **Sprint 2**, because the manifest depends on subscribers' boxes being finalised first.

Q3: How did your simulated sprint align with your plan?

The scope matched the plan, but the timing did not. All 18 planned points in Sprint 1 and all 13 in Sprint 2 were completed, with no scope changes and nothing left over. The timing differs from the plan in two ways. The handout asks for 1-week sprints, but in Jira Sprint 1 runs from 6 to 20 Sep and Sprint 2 from 7 Sep to 4 Oct. And instead of finishing stories one by one across the sprint, all the work was moved to Done on 7 Sep, because the sprints were simulated in a single session.

Q4: What insights did the burndown chart give about your team's capacity?

Both charts show the "sharp drop" pattern the handout describes: a flat line followed by a sudden fall to zero, instead of following the guideline. In a real team this would suggest that work was not updated as it happened or was not broken down well. Here it is because the sprints were simulated, so the charts cannot give a reliable velocity. The only firm figure is that the team delivered 31 story points across two sprints (18 and 13). For a real project, moving each story across the board on the day its work actually finishes would make the burndown a useful measure of capacity.